AMIT KUMAR MISHRA

Python Developer | Machine Learning Engineer

Built end-to-end Python backend and machine learning projects with public source code.

Email | LinkedIn | GitHub | Portfolio

Professional Summary

Physics graduate with hands-on experience building end-to-end Python, machine learning, and AI applications, including backend APIs, data pipelines, model integration, and interactive web interfaces. Experienced with FastAPI, Flask, Python, SQL, React, and production-oriented application development. Seeking Python Developer or Machine Learning Engineer roles.

Technical Skills

Languages: Python, SQL, Java (Basics)
Backend: FastAPI, Flask, REST APIs
Machine Learning: Pandas, NumPy, Scikit-learn, XGBoost, SMOTE
Frontend: React, TypeScript, Tailwind CSS
Databases: Google BigQuery, SQLite
Tools: Git, GitHub, uv, Docker, Linux
Core: Object-oriented programming (OOP), Data Structures & Algorithms (learning)

Projects

MedVerify - AI-Powered Medicine Authenticator

• Built an AI-powered medicine authentication platform using Flask, Groq Llama 3, EasyOCR, OpenCV, and Google BigQuery.
• Designed an offline-first architecture with secure backend validation for low-connectivity environments.
• Implemented REST APIs, OCR pipeline, barcode verification, and production-oriented backend security.
• Repository: github.com/Amit4517187/med-verify-authenticator

Enterprise AI Workspace – Knowledge Platform

- Designing an enterprise knowledge platform that enables users to retrieve trustworthy information from organizational documents using grounded AI responses and source citations.
- Designed a modular FastAPI backend architecture with configuration management, health monitoring, and clear separation of application components for future retrieval and ingestion services.
- Defined an RAG architecture for document ingestion, chunking, hybrid retrieval, evidence selection, citation-backed generation, and access-controlled knowledge retrieval.
- Designed the system around tenant/workspace isolation, document-level permissions, traceability, and evaluation of retrieval and generated responses.
- Repository: github.com/Amit4517187/enterprise-ai-workspace

Credit Underwriting Engine – End to End ML System

- Built an end-to-end credit underwriting system that predicts borrower default risk from structured application and historical credit data.
- Engineered features from application, bureau, and previous-application data and developed a production-oriented risk prediction pipeline.
- Built FastAPI inference services and integrated the prediction workflow with a React frontend for an end-to-end user experience.
- Added business-rule evaluation and LLM-powered explanations to make model predictions more interpretable for users.
- Deployed the application with documented architecture, API contracts, and reproducible project structure.
- Repository: github.com/Amit4517187/credit-engine.

Telecom Customer Churn Prediction

• Built a customer churn prediction pipeline with preprocessing, feature engineering, and model evaluation.
• Repository: github.com/Amit4517187/Telecom-customer-churn-main

Education

B.Sc. Physics | Marwari College, Bhagalpur | 2021-2024 | CGPA: 6.7

Certifications

Deloitte Australia - Data Analytics Job Simulation (Forage)
Google Kaggle - 5-Day AI Agents Intensive
HCL GUVI - AI Impact Summit Buildathon